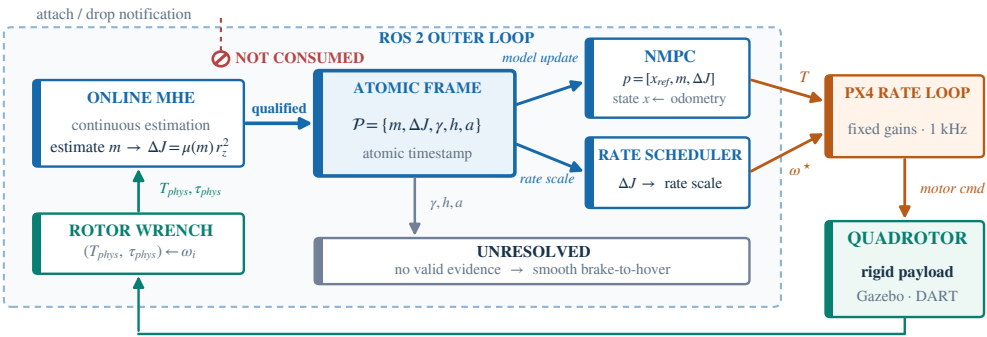
attach / drop notification
ROS 2 OUTER LOOP
NOT CONSUMED
model update
ONLINE MHE
continuous estimation
estimate $m \rightarrow \Delta J = \mu(m)\, r_z^2$
qualified
ATOMIC FRAME
$\mathcal{P} = \{m, \Delta J, \gamma, h, a\}$
atomic timestamp
NMPC
$p = [x_{ref}, m, \Delta J]$
state $x \leftarrow$ odometry
RATE SCHEDULER
$\Delta J \rightarrow$ rate scale
rate scale
T
$\omega^\star$
PX4 RATE LOOP
fixed gains · 1 kHz
motor cmd
QUADROTOR
rigid payload
Gazebo · DART
T_{phys}, τ_{phys}
ROTOR WRENCH
$(T_{phys},\ \tau_{phys}) \leftarrow \omega_i$
γ, h, a
UNRESOLVED
no valid evidence → smooth brake-to-hover
only motor speeds ω_i + odometry x